

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 1)

In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most 90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the most number of students who studied all the three subjects. Then $m + n$ is equal to _____

Q2 - 2024 (04 Apr Shift 2)

Let a relation R on $N \times N$ be defined as: $(x_1, y_1) R (x_2, y_2)$ if and only if $x_1 \leq x_2$ or $y_1 \leq y_2$. Consider the two statements:

(I) R is reflexive but not symmetric.

(II) R is transitive Then which one of the following is true?

(1) Both (I) and (II) are correct.

(2) Only (II) is correct.

(3) Neither (I) nor (II) is correct.

(4) Only (I) is correct.

Q3 - 2024 (05 Apr Shift 2)

Let the set $S = \{2, 4, 8, 16, \dots, 512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to:

(1) 1680

(2) 1640

(3) 1520

(4) 1710

Q4 - 2024 (06 Apr Shift 1)

Questions

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Let the relations R_1 and R_2 on the set $X = \{1, 2, 3, \dots, 20\}$ be given by $R_1 = \{(x, y) : 2x - 3y = 2\}$ and $R_2 = \{(x, y) : -5x + 4y = 0\}$. If M and N be the minimum number of elements required to be added in R_1 and R_2 , respectively, in order to make the relations symmetric, then $M + N$ equals

- (1) 12
- (2) 16
- (3) 8
- (4) 10

Q5 - 2024 (06 Apr Shift 2)

Let $A = \{1, 2, 3, 4, 5\}$. Let R be a relation on A defined by xRy if and only if $4x \leq 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A \times A$ that are required to be added to R to make it a symmetric relation. Then $m + n$ is equal to :

- (1) 25
- (2) 24
- (3) 26
- (4) 23

Q6 - 2024 (08 Apr Shift 2)

Let $A = \{2, 3, 6, 8, 9, 11\}$ and $B = \{1, 4, 5, 10, 15\}$. Let R be a relation on $A \times B$ defined by $(a, b)R(c, d)$ if and only if $3ad - 7bc$ is an even integer. Then the relation R is

- (1) an equivalence relation.
- (2) reflexive and symmetric but not transitive.
- (3) transitive but not symmetric.
- (4) reflexive but not symmetric.

Q7 - 2024 (09 Apr Shift 1)

Questions

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Let $A = \{2, 3, 6, 7\}$ and $B = \{4, 5, 6, 8\}$. Let R be a relation defined on $A \times B$ by $(a_1, b_1) R (a_2, b_2)$ if and only if $a_1 + a_2 = b_1 + b_2$. Then the number of elements in R is _____

Questions

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Answer Key

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Q1 (45)

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Q2 (4)

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Q3 (1)

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Q4 (4)

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Q5 (1)

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Q6 (2)

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Q7 (25)

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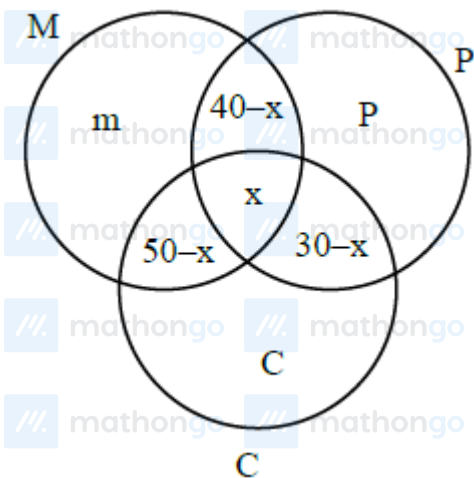
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#MathBoleTohMathonGo

Solutions

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Q1



$$125 \leq m + 90 - x \leq 130$$

$$85 \leq P + 70 - x \leq 95$$

$$75 \leq C + 80 - x \leq 90$$

$$m + P + C + 120 - 2x = 210$$

$$\Rightarrow 15 \leq x \leq 45 \text{ and } 30 - x \geq 0$$

$$\Rightarrow 15 \leq x \leq 30$$

$$30 + 15 = 45$$

Q2

All $((x_1, y_1), (x_1, y_1))$ are in R where

$x_1, y_1 \in \mathbb{N} \therefore R$ is reflexive

$((1, 1), (2, 3)) \in R$ but $((2, 3), (1, 1)) \notin R$

$\therefore R$ is not symmetric

$((2, 4), (3, 3)) \in R$ and $((3, 3), (1, 3)) \in R$ but $((2, 4),$

$(1, 3)) \notin R$

$\therefore R$ is not transitive

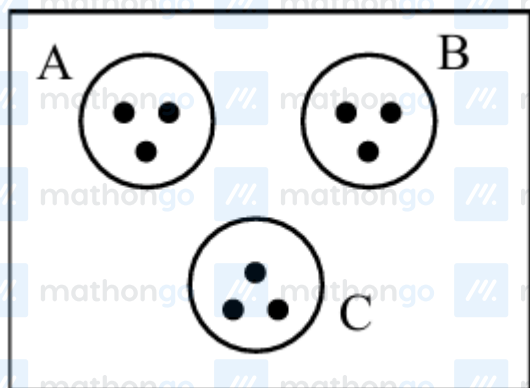
Q3

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Solutions

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$$\frac{9!}{(3!3!3!)3!} \times 3!$$

Q4

$$x = \{1, 2, 3, \dots, 20\}$$

$$R_1 = \{(x, y) : 2x - 3y = 2\}$$

$$R_2 = \{(x, y) : -5x + 4y = 0\}$$

$$R_1 = \{(4, 2), (7, 4), (10, 6), (13, 8), (16, 10), (19, 12)\}$$

$$R_2 = \{(4, 5), (8, 10), (12, 15), (16, 20)\}$$

in R_1 6 element needed

in R_2 4 element needed

So, total $6 + 4 = 10$ element

Q5

$$\text{Given : } 4x \leq 5y$$

then

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5), (4, 4), (4, 5), (5, 4), (5, 5)\}$$

i.e. 16 elements.

$$\text{i.e. } m = 16$$

Now to make R a symmetric relation add

$$\{(2, 1), (3, 2), (4, 3), (3, 1), (4, 2), (5, 3), (4, 1), (5, 2), (5, 1)\}$$

$$\text{i.e. } n = 9$$

$$\text{So } m + n = 25$$

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Solutions

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Q6

$$A = \{2, 3, 6, 8, 9, 11\} \quad (a, b)R(c, d)$$

$$B = \{1, 4, 5, 10, 15\} \quad 3ad - 7bc$$

Reflexive : $(a, b)R(a, b)$

$\Rightarrow 3ab - 7ba = -4ab$ always even so it is reflexive.

Symmetric : If $3ad - 7bc = \text{Even}$

Case-I : odd odd

Case-II : even even

$$(c, d)R(a, b) \Rightarrow 3bc - 3ab$$

Case-I : odd odd

Case-II : even even

so symmetric relation

Transitive :

Set $(3, 4)R(6, 4)$ Satisfy relation

Set $(6, 4)R(3, 1)$ Satisfy relation

but $(3, 4)R(3, 1)$ does not satisfy relation so not transitive.

Q7

$$A = \{2, 3, 6, 7\}$$

$$B = \{2, 5, 6, 8\}$$

$$(a_1, b_1)R(a_2, b_2)$$

$$a_1 + a_2 = b_1 + b_2$$

$$1. (2, 4)R(6, 4) \quad 2. (2, 4)R(7, 5)$$

$$3. (2, 5)R(7, 4) \quad 4. (3, 4)R(6, 5)$$

$$5. (3, 5)R(6, 4) \quad 6. (3, 5)R(7, 5)$$

$$7. (3, 6)R(7, 4) \quad 8. (3, 4)R(7, 6)$$

$$9. (6, 5)R(7, 8) \quad 10. (6, 8)R(7, 5)$$

$$11. (7, 8)R(7, 6) \quad 12. (6, 8)R(6, 4)$$

$$13. (6, 6)R(6, 6)$$

$\times 2$

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Total $24 + 1 = 25$

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